MANAGEMENT OPERATING REVIEW - RECORD 010

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-010

Field Reliability Operating Review for November 2024: purpose and attendance

Standing agenda order was changed so that the constrained function presents first while the room is still fresh. Renata Voss attended as VP People. Notes were circulated on 2024-12-05. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owen Castellanos, VP Customer Success, presented the operations section. The next session is set for 2025-01-09. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

The chair asked each function for the one number it would defend to a customer, which produced a more useful hour than the slides did. Renata Voss attended as VP People. Pre-reads went out ahead of the 2024-12-05 session. Storm closures removed a week of capacity. Two crews were weather-bound for days. The Field Reliability Operating Review convened for November 2024. One function submitted its pre-read late. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

The packet now carries a single page of exceptions, and everything else is available on request rather than presented. The next session is set for 2025-01-09. One function submitted its pre-read late. Storm closures removed a week of capacity. Two crews were weather-bound for days. Renata Voss attended as VP People. Owen Castellanos, VP Customer Success, presented the operations section. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Owners who could not attend sent written positions, and those were read into the record verbatim. Renata Voss attended as VP People. The Field Reliability Operating Review convened for November 2024. Storm closures removed a week of capacity. Two crews were weather-bound for days. Pre-reads went out ahead of the 2024-12-05 session. Owen Castellanos, VP Customer Success, presented the operations section. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Bookings and order intake

The competitive picture shifted, with two losses attributed to a rival that had not appeared in a deal review for three periods. Discount depth peaked on the Tenmile Freight multi-site award. Pinebank Distribution accounted for the largest single order. Storm closures removed a week of capacity. Two crews were weather-bound for days. Open pursuits number 108, up from 111. Win rate held at 31 percent on 108 live opportunities. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Commercial intake for the period came in ahead of the internal plan on unit count and behind it on gross margin, which is the trade the team was asked to make. Pinebank Distribution accounted for the largest single order. Discount depth peaked on the Tenmile Freight multi-site award. Storm closures removed a week of capacity. Two crews were weather-bound for days. The book took 66 units, against 188 last period. Signed volume: 66 units, win rate 31 percent. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Discounting concentrated in one competitive displacement rather than spreading across the book, which the review treated as an acceptable outcome. Pinebank Distribution accounted for the largest single order. Signed volume: 66 units, win rate 31 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. Win rate held at 31 percent on 108 live opportunities. The book took 66 units, against 188 last period. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Late-quarter intake distorted the shape of the funnel again, and the pattern is now visible in three consecutive periods of data. Discount depth peaked on the Tenmile Freight multi-site award. Open pursuits number 108, up from 111. Storm closures removed a week of capacity. Two crews were weather-bound for days. Intake closed at 66 units for November 2024. The book took 66 units, against 188 last period. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

A single reseller relationship produced more qualified volume than the whole outbound programme, which raises an uncomfortable question about channel mix. Intake closed at 66 units for November 2024. Open pursuits number 108, up from 111. Storm closures removed a week of capacity. Two crews were weather-bound for days. Pinebank Distribution accounted for the largest single order. Signed volume: 66 units, win rate 31 percent. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Pricing exceptions ran above the standing threshold this period, mostly on multi-site commitments where the customer wanted a single unit price. Win rate held at 31 percent on 108 live opportunities. The book took 66 units, against 188 last period. Storm closures removed a week of capacity. Two crews were weather-bound for days. Intake closed at 66 units for November 2024. Open pursuits number 108, up from 111. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Installation crew loading, November 2024

Crew	Home region	Certified leads	Field engineers	Utilisation %	Weeks committed forward
Crew Alpha	Southeast	2	5	92	7
Crew Bravo	Northeast	2	3	87	15
Crew Charlie	Ontario	2	8	94	5
Crew Delta	Channel and reseller	2	3	89	14
Crew Echo	West	1	8	96	10
Crew Foxtrot	Southwest	1	6	85	13
Crew Gulf	Great Lakes	1	8	93	5
Crew Hotel	Southeast	1	3	88	11
Contract crew Northfield	Northeast	1	8	95	5
Contract crew Verrado	Ontario	1	4	86	14
Contract crew Ellingham	Channel and reseller	1	8	91	5
Commissioning bench	West	1	5	84	13
Crew Alpha (night rotation)	Southwest	1	5	92	9
Crew Bravo (night rotation)	Great Lakes	1	3	87	17
Crew Charlie (night rotation)	Southeast	1	8	94	7
Crew Delta (night rotation)	Northeast	1	3	89	11
Crew Echo (night rotation)	Ontario	1	8	96	7
Crew Foxtrot (night rotation)	Channel and reseller	1	6	85	15

Purchasing and lead times

Where tooling belongs to the supplier, the exit cost is real and it is not written down anywhere. Coverage weeks on long-lead parts were extended. On-time delivery was 87 percent, from 90 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. Penrose Castings missed two consecutive commit dates. Expedite activity rose with the build plan of 66. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Lead-time creep is the pattern to watch, because quoted lead times moved without any supplier formally notifying a change. Calder Battery Works improved after a site visit. On-time delivery was 87 percent, from 90 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. Penrose Castings missed two consecutive commit dates. Owen Castellanos issued the first supplier scorecards. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Purchase order acknowledgements are now chased on a fixed cadence, which sounds trivial and caught two silent date pushes. Penrose Castings missed two consecutive commit dates. Coverage weeks on long-lead parts were extended. Storm closures removed a week of capacity. Two crews were weather-bound for days. On-time delivery was 87 percent, from 90 percent. Supplier on-time delivery came in at 87 percent. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Forecast sharing with the top five suppliers began this period, and two of them immediately improved their commit dates. Penrose Castings missed two consecutive commit dates. Supplier on-time delivery came in at 87 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. On-time delivery was 87 percent, from 90 percent. Coverage weeks on long-lead parts were extended. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Dual sourcing costs more per unit during the overlap period, and the review accepted that cost with its eyes open. Supplier on-time delivery came in at 87 percent. Penrose Castings missed two consecutive commit dates. Storm closures removed a week of capacity. Two crews were weather-bound for days. On-time delivery was 87 percent, from 90 percent. Calder Battery Works improved after a site visit. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Tooling ownership is unclear on four part numbers, and the review asked for the contracts to be read rather than remembered. Penrose Castings missed two consecutive commit dates. Expedite activity rose with the build plan of 66. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owen Castellanos issued the first supplier scorecards. Supplier on-time delivery came in at 87 percent. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Build attainment by week and line, November 2024

Week	Product line	Plan units	Actual units	Variance units	Primary constraint
W41	HS-200 tote handler	8	6	-2	operator absence
W41	HS-400 pallet mover	4	3	-1	test cell queue
W42	HS-200 tote handler	5	4	-1	none
W42	HS-400 pallet mover	3	2	-1	fixture failure
W43	HS-200 tote handler	3	4	1	drive unit shortage
W43	HS-400 pallet mover	1	2	1	sensor head rework
W44	HS-200 tote handler	3	4	1	test cell queue
W44	HS-400 pallet mover	1	3	2	harness shortage
W45	HS-200 tote handler	3	4	1	fixture failure
W45	HS-400 pallet mover	2	3	1	plan revision mid-week
W46	HS-200 tote handler	3	5	2	sensor head rework
W46	HS-400 pallet mover	2	3	1	operator absence
W47	HS-200 tote handler	4	2	-2	harness shortage
W47	HS-400 pallet mover	2	1	-1	none
W48	HS-200 tote handler	4	3	-1	plan revision mid-week
W48	HS-400 pallet mover	2	1	-1	drive unit shortage
W49	HS-200 tote handler	4	3	-1	operator absence
W49	HS-400 pallet mover	3	1	-2	test cell queue

Work in process and buffers

Consignment terms with two suppliers would change the cash picture more than any other inventory action available. Kit shortages were concentrated in low-value parts. Renata Voss owns the disposition decision. Storm closures removed a week of capacity. Two crews were weather-bound for days. Buffer stock on the drive unit was raised deliberately. Floor space in staging is now the binding limit. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Inventory turns are a poor metric while the company is scaling, so the review looked at coverage weeks by commodity instead. Renata Voss owns the disposition decision. Work in process closed at 31 units. Storm closures removed a week of capacity. Two crews were weather-bound for days. Kit shortages were concentrated in low-value parts. Buffer stock on the drive unit was raised deliberately. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Receiving is holding returned parts that could be reworked, and nobody owns the decision to rework them. Kit shortages were concentrated in low-value parts. Excess exposure sits in the superseded sensor head. Storm closures removed a week of capacity. Two crews were weather-bound for days. Buffer stock on the drive unit was raised deliberately. Work in process closed at 31 units. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Inventory is growing faster than build volume, and the growth is in long-lead components rather than in finished goods. Work in process moved from 33 to 31 units. Excess exposure sits in the superseded sensor head. Storm closures removed a week of capacity. Two crews were weather-bound for days. Kit shortages were concentrated in low-value parts. Renata Voss owns the disposition decision. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Finished-goods staging is now constrained by floor space, which is an odd problem to have while missing the build plan. Renata Voss owns the disposition decision. Work in process closed at 31 units. Storm closures removed a week of capacity. Two crews were weather-bound for days. Buffer stock on the drive unit was raised deliberately. Work in process moved from 33 to 31 units. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Cycle counting accuracy improved after the bin relabelling, and the count variance is now small enough to trust the system quantity. Work in process moved from 33 to 31 units. Renata Voss owns the disposition decision. Storm closures removed a week of capacity. Two crews were weather-bound for days. Floor space in staging is now the binding limit. Buffer stock on the drive unit was raised deliberately. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Service performance across the fleet

Software releases are now staged across sites rather than pushed at once, after a release regressed docking behaviour at two accounts. Kennesaw remains the weakest site on interventions. Interventions spaced 318 hours apart on the fleet median. Storm closures removed a week of capacity. Two crews were weather-bound for days. Fleet uptime averaged 96.7 percent. Uptime reached 96.7 percent, from 97.1 percent. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Remote diagnostics closed more incidents without a visit than in any prior period, which is where the cost curve bends. Kennesaw remains the weakest site on interventions. Pinebank Distribution measures uptime on a stricter basis. Storm closures removed a week of capacity. Two crews were weather-bound for days. Mean time between interventions was 318 hours. Uptime reached 96.7 percent, from 97.1 percent. The HS-200 tote handler took most of the test cell time.

Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Reliability engineering split the top failure modes by build vintage and found that one of them is entirely a legacy population problem. Mean time between interventions was 318 hours. Interventions spaced 318 hours apart on the fleet median. Storm closures removed a week of capacity. Two crews were weather-bound for days. Griffin Tasse leads the joint remediation plan. Pinebank Distribution measures uptime on a stricter basis. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Fleet uptime across all live sites improved, and the improvement is broad rather than driven by one recovering account. Mean time between interventions was 318 hours. Interventions spaced 318 hours apart on the fleet median. Storm closures removed a week of capacity. Two crews were weather-bound for days. Kennesaw remains the weakest site on interventions. Uptime reached 96.7 percent, from 97.1 percent. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Site-level variation in reliability is wider than the fleet average suggests, and the worst site drags the contractual conversation. Mean time between interventions was 318 hours. Kennesaw remains the weakest site on interventions. Storm closures removed a week of capacity. Two crews were weather-bound for days. Interventions spaced 318 hours apart on the fleet median. Fleet uptime averaged 96.7 percent. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

The remediation plan with the worst-performing account is joint, with obligations written on both sides and a fixed measurement window. Pinebank Distribution measures uptime on a stricter basis. Mean time between interventions was 318 hours. Storm closures removed a week of capacity. Two crews were weather-bound for days. Fleet uptime averaged 96.7 percent. Uptime reached 96.7 percent, from 97.1 percent. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Assembly cell performance by shift, November 2024

Cell	Shift	Units processed	First-pass yield %	Rework hours	Scrap USD thousands
Frame and drive assembly	first shift	14	93	4	7
Sensor head assembly	first shift	12	85	100	1
Harness and electrical	first shift	12	91	15	2
Final assembly	first shift	11	92	50	2
Functional test	first shift	9	90	25	1
Pack and ship	first shift	12	93	60	2
Frame and drive assembly	second shift	8	90	5	4
Sensor head assembly	second shift	6	83	105	1
Harness and electrical	second shift	6	89	20	1
Final assembly	second shift	6	90	55	1
Functional test	second shift	5	90	10	1
Pack and ship	second shift	7	88	35	1
Frame and drive assembly	weekend cover	2	93	4	1
Sensor head assembly	weekend cover	2	85	100	1
Harness and electrical	weekend cover	2	91	15	1
Final assembly	weekend cover	2	92	50	1
Functional test	weekend cover	1	90	25	1
Pack and ship	weekend cover	2	93	60	1

Spares, depots and warranty

Depot placement was reconsidered, because half of the courier spend goes to sites within a day's drive of a hub. Return material flow remains slow. Ingrid Solberg approved the critical spares increase. Storm closures removed a week of capacity. Two crews were weather-bound for days. Fill rate ran 90 percent against 94 percent. Warranty cost per fleet

unit is falling. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

The recall-style replacement of affected sensor heads is proceeding on a planned schedule rather than on failure. Warranty cost per fleet unit is falling. Return material flow remains slow. Storm closures removed a week of capacity. Two crews were weather-bound for days. Depot fill rate was 90 percent. Ingrid Solberg approved the critical spares increase. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Service parts pricing has never been reviewed, and two parts are being sold below their landed cost. Fill rate ran 90 percent against 94 percent. Elk Grove took the planned sensor head replacements. Storm closures removed a week of capacity. Two crews were weather-bound for days. Vehicle stock is masking the true fill rate. Warranty cost per fleet unit is falling. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Obsolescence on the first-generation controller will force a spares buy decision before the end of next year. Elk Grove took the planned sensor head replacements. Warranty cost per fleet unit is falling. Storm closures removed a week of capacity. Two crews were weather-bound for days. Fill rate ran 90 percent against 94 percent. Vehicle stock is masking the true fill rate. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

First-generation controllers are heading for obsolescence, and the spares buy has to be decided before the line stops making them. Warranty cost per fleet unit is falling. Ingrid Solberg approved the critical spares increase. Storm closures removed a week of capacity. Two crews were weather-bound for days. Depot fill rate was 90 percent. Elk Grove took the planned sensor head replacements. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Warranty accrual assumptions were set before the yield excursion and have not been revisited since. Fill rate ran 90 percent against 94 percent. Warranty cost per fleet unit is falling. Storm closures removed a week of capacity. Two crews were weather-bound for days. Return material flow remains slow. Vehicle stock is masking the true fill rate. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Carried constraints

Constraint under review: sensor assembly yield, at the field escapes stage as of this session. Nineteen sensor heads built during the affected window reached customer sites, and the service team is replacing them on a planned schedule rather than waiting for failures. Days outstanding 61, Pinebank Distribution the largest balance. Queue depth 19 sites, Kennesaw at the front.

Constraint under review: large-account uptime, at the intervention analysis stage as of this session. Mean time between interventions at Cedar Bluff is two hundred and ten hours against a fleet median of three hundred and forty, and most of the difference is floor marking, not the robots. Uptime 96.7 percent, interventions every 318 hours. Utilisation 90 percent, Crew Alpha fully committed.

Purchase order acknowledgements are now chased on a fixed cadence, which sounds trivial and caught two silent date pushes. Owen Castellanos issued the first supplier scorecards. Calder Battery Works improved after a site visit. Storm closures removed a week of capacity. Two crews were weather-bound for days. Penrose Castings missed two consecutive commit dates. On-time delivery was 87 percent, from 90 percent. Queue depth 19 sites, Kennesaw at the front. Crew Alpha closed Ashland Park without a second visit. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Attrition 10 percent annualised in Northeast.

Yield excursion and containment log, November 2024

Week	Cell	First-pass yield %	Units affected	Containment in force	Disposition
W41	Harness and electrical	84	5	full inspection reinstated	rework and release

Week	Cell	First-pass yield %	Units affected	Containment in force	Disposition
W42	Final assembly	92	10	standard sampling	scrap
W43	Sensor head assembly	87	9	100 percent bond-line inspection	use as is with concession
W44	Harness and electrical	94	7	sample one in ten	return to supplier
W45	Final assembly	89	6	shipments held	hold pending study
W46	Sensor head assembly	96	9	inspection lifted, trigger armed	rework and release
W47	Harness and electrical	85	8	full inspection reinstated	scrap
W48	Final assembly	93	5	standard sampling	use as is with concession
W49	Sensor head assembly	88	10	100 percent bond-line inspection	return to supplier
W50	Harness and electrical	95	6	sample one in ten	hold pending study
W51	Final assembly	86	7	shipments held	rework and release
W52	Sensor head assembly	91	8	inspection lifted, trigger armed	scrap
W01	Harness and electrical	84	5	full inspection reinstated	use as is with concession
W02	Final assembly	92	10	standard sampling	return to supplier
W03	Sensor head assembly	87	9	100 percent bond-line inspection	hold pending study
W04	Harness and electrical	94	7	sample one in ten	rework and release
W05	Final assembly	89	6	shipments held	scrap
W06	Sensor head assembly	96	9	inspection lifted, trigger armed	use as is with concession
W07	Harness and electrical	85	8	full inspection reinstated	return to supplier
W08	Final assembly	93	5	standard sampling	hold pending study

People and hiring

Contractor-to-employee conversion is the cheapest source of certified crew, and the process for it does not exist yet. Requisitions fell from 23 to 21. Time to fill improved from 77 to 72 days. Storm closures removed a week of capacity. Two crews were weather-bound for days. Median time to fill was 72 days. Annualised attrition sits at 10 percent. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Headcount against plan closed part of its gap, and the remaining gap is concentrated in the two functions that gate deployment capacity. Open requisitions number 21. Time to fill improved from 77 to 72 days. Storm closures removed a week of capacity. Two crews were weather-bound for days. Median time to fill was 72 days. Requisitions fell from 23 to 21. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Open requisitions were re-ranked against the operating plan, and four were closed rather than left to age quietly. Time to fill improved from 77 to 72 days. Open requisitions number 21. Storm closures removed a week of capacity. Two crews were weather-bound for days. Headcount stood at 172 against a plan of 181. Median time to fill was 72 days. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Manager span of control has grown past what the review considers sustainable in two functions. Median time to fill was 72 days. Annualised attrition sits at 10 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. The headcount gap is 9 roles. Headcount stood at 172 against a plan of 181. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

A hiring pause was applied selectively, protecting customer-facing roles and holding back internal ones. Median time to fill was 72 days. Time to fill improved from 77 to 72 days. Storm closures removed a week of capacity. Two crews were weather-bound for days. The headcount gap is 9 roles. Open requisitions number 21. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Engagement survey results split by tenure rather than by function, with the newest cohort least positive about clarity of direction. Median time to fill was 72 days. Annualised attrition sits at 10 percent. Storm closures removed a week of capacity. Two crews were weather-bound for days. Open requisitions number 21. The headcount gap is 9 roles. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Invoicing and days outstanding

Customer-side accounts payable portals are a real source of delay, and three accounts require a different submission path. Tenmile Freight requires a separate submission path. Two receivables aged past ninety days. Storm closures removed a week of capacity. Two crews were weather-bound for days. Days sales outstanding closed at 61 days. Receipts of 2530 thousand outpaced invoicing. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Progress billing on long deployments would smooth the cash profile, and the commercial team has resisted it for good reasons. Days sales outstanding closed at 61 days. Tenmile Freight requires a separate submission path. Storm closures removed a week of capacity. Two crews were weather-bound for days. Receipts of 2530 thousand outpaced invoicing. Outstanding days improved from 64 to 61. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Aged balances were read out account by account, which took ten minutes and closed two of them the same week. Receipts of 2530 thousand outpaced invoicing. Nadia Oyelaran moved collections calls to the account teams. Storm closures removed a week of capacity. Two crews were weather-bound for days. Two receivables aged past ninety days. Collections totalled 2530 thousand dollars. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Days sales outstanding is the metric under active management, and the improvement came from the invoicing process rather than from customer behaviour. Receipts of 2530 thousand outpaced invoicing. Collections totalled 2530 thousand dollars. Storm closures removed a week of capacity. Two crews were weather-bound for days. Tenmile Freight requires a separate submission path. Two receivables aged past ninety days. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Two large receivables aged past ninety days, and both trace to a purchase order number mismatch rather than to unwillingness to pay. Tenmile Freight requires a separate submission path. Two receivables aged past ninety days. Storm closures removed a week of capacity. Two crews were weather-bound for days. Receipts of 2530 thousand outpaced invoicing. Nadia Oyelaran moved collections calls to the account teams. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Payment terms granted in commercial negotiation were never visible to the finance team until invoicing, and that has been fixed. Receipts of 2530 thousand outpaced invoicing. Outstanding days improved from 64 to 61. Storm closures removed a week of capacity. Two crews were weather-bound for days. Two receivables aged past ninety days. Tenmile Freight requires a separate submission path. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Supplier delivery and lead time, November 2024

Supplier	Commodity	On-time delivery %	Quoted lead weeks	Actual lead weeks	Watch status
Kestrel Drive Systems	Drive units	82	8	16	on scorecard review
Brightwater Motion	Lidar assemblies	90	10	10	no action
Halverson Optics	Battery packs	85	12	18	watch
Penrose Castings	Castings	92	14	14	escalated
Ironbark Sheet Metal	Sheet metal	83	15	18	cleared
Calder Battery Works	Wire harness	88	11	11	second source in qualification
Tessellate Controls	Controllers	81	12	17	on scorecard review

Supplier	Commodity	On-time delivery %	Quoted lead weeks	Actual lead weeks	Watch status
Ryemoor Wire Harness	Caster and wheel sets	89	13	13	no action
Kestrel Drive Systems	Lidar assemblies	84	14	15	watch
Brightwater Motion	Battery packs	91	15	15	escalated
Halverson Optics	Castings	86	11	13	cleared
Penrose Castings	Sheet metal	93	12	12	second source in qualification
Ironbark Sheet Metal	Wire harness	82	13	17	on scorecard review
Calder Battery Works	Controllers	90	14	14	no action
Tessellate Controls	Caster and wheel sets	85	15	21	watch
Ryemoor Wire Harness	Drive units	92	11	11	escalated

Financial posture

Cash discipline this year came two-thirds from collections and one-third from deferred capital, which is a better mix than cutting people. Test-cell capital was deferred again. Owen Castellanos holds the trigger list. Storm closures removed a week of capacity. Two crews were weather-bound for days. Burn for the period was 2910 thousand dollars. Runway moved from 19 to 18 months. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Runway is not a single number, and presenting it as one has caused more confusion than it has prevented. Owen Castellanos holds the trigger list. Spending ran 2910 thousand against 8520 thousand. Storm closures removed a week of capacity. Two crews were weather-bound for days. Capital spend was 360 thousand dollars. Test-cell capital was deferred again. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Each scenario was read against a named observable, so nobody has to interpret whether a trigger has fired. Test-cell capital was deferred again. Spending ran 2910 thousand against 8520 thousand. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owen Castellanos holds the trigger list. Runway moved from 19 to 18 months. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Capital spending on test cells was deferred, which protects cash now and constrains build capacity later. Runway stands at 18 months on the middle case. Owen Castellanos holds the trigger list. Storm closures removed a week of capacity. Two crews were weather-bound for days. Runway moved from 19 to 18 months. Spending ran 2910 thousand against 8520 thousand. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

The downside trigger was armed after two large bookings left the quarter, and nine requisitions were paused within a week. Owen Castellanos holds the trigger list. Runway stands at 18 months on the middle case. Storm closures removed a week of capacity. Two crews were weather-bound for days. Spending ran 2910 thousand against 8520 thousand. Test-cell capital was deferred again. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Fixed cost per period has grown with facilities and tooling commitments that cannot be unwound inside a quarter. Capital spend was 360 thousand dollars. Runway stands at 18 months on the middle case. Storm closures removed a week of capacity. Two crews were weather-bound for days. Spending ran 2910 thousand against 8520 thousand. Burn for the period was 2910 thousand dollars. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Expedite and premium freight log, November 2024

Part number	Supplier	Reason	Freight mode	Added cost USD thousands
4110-1	Halverson Optics	yield loss replacement	courier	17
4115-2	Penrose Castings	spares depot backorder	expedited road	11
4120-3	Ironbark Sheet Metal	engineering change cut-in	air	21
4125-4	Calder Battery Works	customer date brought	standard road with	13

Part number	Supplier	Reason	Freight mode	Added cost USD thousands
		forward	premium	
4130-5	Tessellate Controls	build plan pull-in	courier	15
4135-6	Ryemoor Wire Harness	supplier missed commit date	expedited road	17
4140-7	Kestrel Drive Systems	yield loss replacement	air	11
4145-8	Brightwater Motion	spares depot backorder	standard road with premium	21
4150-0	Halverson Optics	engineering change cut-in	courier	19
4155-1	Penrose Castings	customer date brought forward	expedited road	15
4160-2	Ironbark Sheet Metal	build plan pull-in	air	13
4165-3	Calder Battery Works	supplier missed commit date	standard road with premium	19
4170-4	Tessellate Controls	yield loss replacement	courier	17
4175-5	Ryemoor Wire Harness	spares depot backorder	expedited road	11
4180-6	Kestrel Drive Systems	engineering change cut-in	air	21
4185-7	Brightwater Motion	customer date brought forward	standard road with premium	13
4190-8	Halverson Optics	build plan pull-in	courier	15

Engineering roadmap

Simulation coverage before release improved, and the last two releases produced fewer field regressions as a result. Adapter work crowded out platform work again. Renata Voss brought the decision to stop one feature. Storm closures removed a week of capacity. Two crews were weather-bound for days. Two roadmap checkpoints cleared; one slipped. Release staging across sites is now standard. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Stopping work is harder than starting it, and this session stopped one thing on purpose. Release staging across sites is now standard. Two roadmap checkpoints cleared; one slipped. Storm closures removed a week of capacity. Two crews were weather-bound for days. Adapter work crowded out platform work again. Field data now sets the reliability backlog order. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Roadmap checkpoints were reviewed against the commitments made to customers rather than against the internal plan, which is a harder test. Field data now sets the reliability backlog order. Documentation lags the current release. Storm closures removed a week of capacity. Two crews were weather-bound for days. Adapter work crowded out platform work again. Renata Voss brought the decision to stop one feature. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Hardware and software roadmaps are coupled more tightly than the planning process admits, and one dependency was found late. Documentation lags the current release. Renata Voss brought the decision to stop one feature. Storm closures removed a week of capacity. Two crews were weather-bound for days. Release staging across sites is now standard. Field data now sets the reliability backlog order. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

The next controller revision closes an obsolescence exposure and requires a re-validation cycle that has to be planned now. Adapter work crowded out platform work again. Field data now sets the reliability backlog order. Storm closures removed a week of capacity. Two crews were weather-bound for days. Documentation lags the current release. The HS-200 tote handler absorbed most of the engineering effort. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Documentation and training material lag every release, and the lag is now a deployment risk rather than a nuisance. Renata Voss brought the decision to stop one feature. Field data now sets the reliability backlog order. Storm closures removed a week of capacity. Two crews were weather-bound for days. Adapter work crowded out platform work again. The HS-200 tote handler absorbed most of the engineering effort. Scrap 30 thousand, Calder Battery Works contributing.

Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Unresolved positions

Whether a deferral saves money depends on which function is counting, and both counts were tabled. The disagreement is recorded rather than resolved. No consensus was reached; the VP Customer Success decided. Storm closures removed a week of capacity. Two crews were weather-bound for days. A further test was added because of the dissent. The minority view is preserved in the record. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

Disagreement was recorded rather than smoothed over, because the two positions rest on different readings of the same data. No consensus was reached; the VP Customer Success decided. The objection was noted against the decision. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owen Castellanos and Renata Voss read the data differently. Nadia Oyelaran asked for the decision to be revisited. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Finance and deployment disagreed about whether contract labour should be treated as a fixed or variable commitment. A further test was added because of the dissent. No consensus was reached; the VP Customer Success decided. Storm closures removed a week of capacity. Two crews were weather-bound for days. The disagreement is recorded rather than resolved. Nadia Oyelaran asked for the decision to be revisited. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

Two functions read the reliability data differently, and the review agreed to a single definition before arguing again. Nadia Oyelaran asked for the decision to be revisited. A further test was added because of the dissent. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owen Castellanos and Renata Voss read the data differently. The disagreement is recorded rather than resolved. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Product and sales disagreed on whether the pilot feature should be committed to a customer date, and the answer was no. Nadia Oyelaran asked for the decision to be revisited. The minority view is preserved in the record. Storm closures removed a week of capacity. Two crews were weather-bound for days. The objection was noted against the decision. The disagreement is recorded rather than resolved. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Days outstanding 61, Pinebank Distribution the largest balance. Bookings 66 units across 108 pursuits. Crew Alpha closed Ashland Park without a second visit.

Two owners left the room unconvinced, and the record says so rather than pretending otherwise. The objection was noted against the decision. Owen Castellanos and Renata Voss read the data differently. Storm closures removed a week of capacity. Two crews were weather-bound for days. Nadia Oyelaran asked for the decision to be revisited. No consensus was reached; the VP Customer Success decided. Scrap 30 thousand, Calder Battery Works contributing. Functional test ran 90 percent first pass. Headcount 172 of 181, Owen Castellanos reporting. Slipped deals 9, including Tenmile Freight.

Second-source qualification progress reviewed in November 2024

Qualification step	Candidate source	Owner	Planned date	Actual date	Result
Request for quotation issued	Brightwater Motion drive unit	Hollis Bramwell	2024-09-26	2024-09-24	complete
Sample order placed	Brightwater Motion drive unit	Tobias Nkemdirim	2024-10-10	2024-10-11	in progress
First article dimensional inspection	Brightwater Motion drive unit	Emil Farkas	2024-10-24	2024-10-28	pass
Thermal soak test	Brightwater Motion drive unit	Hollis Bramwell	2024-11-07	2024-11-14	fail, two of eight samples
Endurance run to two thousand hours	Brightwater Motion drive unit	Hollis Bramwell	2024-11-21	2024-11-20	pass with observation

Qualification step	Candidate source	Owner	Planned date	Actual date	Result
Process audit at supplier	Brightwater Motion drive unit	Emil Farkas	2024-12-05	2024-12-07	retest scheduled
Production part approval	Brightwater Motion drive unit	Emil Farkas	2024-12-19	2024-12-24	complete
First production lot	Brightwater Motion drive unit	Hollis Bramwell	2025-01-02	2025-01-10	in progress
Request for quotation issued	Penrose Castings frame	Emil Farkas	2024-10-01	2024-10-01	pass
Sample order placed	Penrose Castings frame	Hollis Bramwell	2024-10-15	2024-10-18	fail, two of eight samples
First article dimensional inspection	Penrose Castings frame	Tobias Nkemdirim	2024-10-29	2024-11-04	pass with observation
Thermal soak test	Penrose Castings frame	Emil Farkas	2024-11-12	2024-11-10	retest scheduled
Endurance run to two thousand hours	Penrose Castings frame	Hollis Bramwell	2024-11-26	2024-11-27	complete
Process audit at supplier	Penrose Castings frame	Hollis Bramwell	2024-12-10	2024-12-14	in progress
Production part approval	Penrose Castings frame	Hollis Bramwell	2024-12-24	2024-12-31	pass
First production lot	Penrose Castings frame	Emil Farkas	2025-01-07	2025-01-06	fail, two of eight samples

Decisions, owners and dates recorded in November 2024

One decision was escalated out of this forum because it commits capital beyond the delegated threshold. Owners were named and dates set against 2025-01-09. One item closed with measured evidence. Storm closures removed a week of capacity. Two crews were weather-bound for days. Two due dates were changed in the room. The oldest item dates from two sessions back. Attrition 10 percent annualised in Northeast. Win rate 31 percent, best in Northeast. Slipped deals 9, including Tenmile Freight. Yield 90 percent, scrap 30 thousand in Functional test.

The oldest open item on the list dates from three sessions ago, and its owner was asked to either finish it or kill it. Three items carried forward into 2025-01-09. One item closed with measured evidence. Storm closures removed a week of capacity. Two crews were weather-bound for days. Ingrid Solberg inherited an item after a role change. Griffin Tasse, Head of Product, took the largest open item. The HS-200 tote handler took most of the test cell time. Crew Alpha closed Ashland Park without a second visit. Functional test ran 90 percent first pass. Capital spend 360 thousand this period.

Nothing was recorded as decided unless somebody in the room owned it and a date was attached to it. Griffin Tasse, Head of Product, took the largest open item. Three items carried forward into 2025-01-09. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owners were named and dates set against 2025-01-09. The oldest item dates from two sessions back. Uptime 96.7 percent, interventions every 318 hours. Requisitions open 21, Renata Voss holding the list. Queue depth 19 sites, Kennesaw at the front. Griffin Tasse owns the Wire harness recovery.

An item was killed rather than rolled a fourth time, and the reasoning is in the record. Two due dates were changed in the room. One item closed with measured evidence. Storm closures removed a week of capacity. Two crews were weather-bound for days. Owners were named and dates set against 2025-01-09. Griffin Tasse, Head of Product, took the largest open item. Slipped deals 9, including Tenmile Freight. Build plan 66, actual 63, Wire harness short. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Queue depth 19 sites, Kennesaw at the front.

Action items reviewed and recorded, November 2024

Item	Description	Owner	Opened	Due	Status	Closing evidence
OR-0409	Agree joint remediation plan with the Cedar Bluff account	Sunil Achterberg	2024-06-06	2024-12-27	Open, date revised	partial
OR-1006	Replace affected sensor heads at live sites on schedule	Sunil Achterberg	2024-12-05	2025-01-19	Open	none yet
OR-1020	Re-cut the stage definitions in the pipeline report	Marguerite Do	2024-11-26	2024-12-16	Closed	report filed
OR-1023	Publish the	Hollis Bramwell	2024-11-22	2024-12-22	Open	outstanding

Item	Description	Owner	Opened	Due	Status	Closing evidence
	supplier scorecard with weighted criteria					
OR-1026	Move collections calls to the account teams	Lucia Mbeki-Hart	2024-11-18	2024-12-28	In progress	outstanding
OR-1029	Set a plan freeze window inside the build month	Emil Farkas	2024-11-14	2025-01-03	Closed	measured result attached
OR-1032	Reconcile the two uptime definitions with the customer	Owen Castellanos	2024-11-10	2025-01-09	Open	outstanding
OR-1035	Cost the second-shift ramp against overtime	Marcus Peil	2024-11-06	2025-01-15	In progress	outstanding
OR-1038	Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2024-11-02	2025-01-21	Closed	report filed
OR-1041	Complete process capability study on the bond operation	Emil Farkas	2024-10-29	2025-01-27	Open	outstanding
OR-1044	Draft deposit terms for multi-site commitments	Ingrid Solberg	2024-10-25	2025-02-02	In progress	outstanding
OR-1047	Bring deal-level notes for the top ten pursuits	Marguerite Do	2024-10-21	2025-02-08	Closed	report filed
OR-1050	Refresh compensation bands for field engineering	Renata Voss	2024-10-17	2025-02-14	Open	outstanding
OR-1053	Measure onboarding time to productivity for crews	Renata Voss	2024-10-13	2025-02-20	In progress	outstanding
OR-1056	Stage software releases across sites rather than at once	Griffin Tasse	2024-10-09	2025-02-26	Closed	memo circulated
OR-1059	Reissue the deployment playbook network section	Priya Ramanathan	2024-10-05	2025-03-04	Open	outstanding